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Course Description

8th Grade STEM is an introduction to Chemistry and Physics. The course is focused on ways to unite our world view as citizens of the planet, nurture our understanding of how chemistry and physics are interrelated and inspire the next generation of scientists and engineers.

The year begins with a short introduction to the Scientific Method and the Engineering Design Process as well as the principal tools and techniques of a scientist and engineer. Then it's time to launch into the study of matter, the elements, the principles of chemistry, and finally an investigation of polymers and the impact they are having on our environment. After the Winter Break, our third unit will find us going back in time, to discover how the science of Physics developed. We will investigate the laws of motion developed by Sir Isaac Newton and build a few simple machines to throw things around the lab. After Spring Break, we will conclude the year with an investigation of Faraday's Fundamental Forces, electricity and magnetism, and how these two fundamental forces act together.

2026-2027 Scope and Sequence:

August-June

All Unit Dates are *projected* and may be subject to change.

Unit 1: I am a Scientist and an Engineer

3 Aug – 28 Aug

Key Concept: Systems

Related Concept(s): Patterns, Form, Function

Global Context: Science and Technical Innovation

Inquiry statement: The Scientific Method and the Engineering Design Process allow us to investigate the form and function of patterns observed in natural systems or designs.

Inquiry questions:

Factual: How do scientists and engineers use the Scientific Method and the Engineering Design Process?

Conceptual: How does experimental science and engineering expand our body of knowledge?

Debatable: Can the Scientific Method or the Engineering Design Process be used to answer any question or solve any problem?

Content

- **Know:** Scientific Method, Engineering Design Process, Properties of Matter
- **Understand:** How the Scientific Method/Engineering Design Process solves problems. Review the linkage between Mass, Volume, and Density. Investigate heat energy.
- **Do:** Lab Safety, Scientific Method/Experimental Cycle, variables and hypothesis formulation, Lab report format, intro to probe ware.

Summative Assessment Tasks

- Properties of Matter Lab (Crit. B,C)

Unit 2: A Period Matters!

1 Sep – 18 Dec

Key Concept: Change

Related Concept(s): Models, Patterns, Consequences

Global Context: Science and Technical Innovation

Inquiry statement: Scientists use models to understand the patterns of change in matter and the consequences of science and technology on our world.

Inquiry questions:

Factual: How are elements organized in the periodic table?

Conceptual: Why is Conservation of Matter so important in chemistry?

Debatable: Should society have control/approval over scientific research?

Content

- **Know:** Structure and states of matter, elements of the Periodic Table, atomic models, types of chemical reactions, field study methods.
- **Understand:** How the atomic theory of interactions is the basis for chemical reactions. How to translate theory into practical application (stoichiometry).
- **Do:** Metrics Lab, Lab Skills Assessment, Chemical Reactions Lab, Acids/Bases Lab, Nuts and Bolts Lab, Stain Lab, Polymer Lab.

Summative Assessment Tasks

Element Babybook (Crit. A/D)

Berth Project (Crit A/D)

Stain Lab (Crit. B/C)

Unit 3: Forces and Motion

7 Jan - 12 Mar

Key Concept: Change

Related Concept(s): Transformation, Movement, Energy

Global Context: Orientation in Space and Time

Inquiry statement: Observing changes in movement allows scientists to detect and understand the transformation of energy.

Inquiry questions:

Factual: Why are "unbalanced" forces so important in the understanding of physics?

Conceptual: How are the Laws of Conservation of Motion and Energy related?

Debatable: Must religion and science always be at odds with one another?

Content

- **Know:** Newton's Laws of Motion, What is velocity and acceleration, Newton's Universal Law of Gravity.
- **Understand:** How unbalanced forces result in the changes we observe in physics.
- **Do:** Simple Machines Lab, Mover Lab, Velocity Lab, Acceleration Lab, Projectile Lab.

Summative Assessment Tasks

The Apprentices Project (Crit. A/B/CD)

Unit 4: Faraday's Fundamental Forces.

30 Mar – 03 Jun

Key Concept: Systems

Related Concepts(s): Balance, Energy, Consequences

Global Context: Globalization and Sustainability

Inquiry statement Understanding the balance between the uses of electronics, AI and robotics and the potential consequences will allow us to develop sustainable global systems.

Inquiry questions:

Factual: How is magnetism related to the production of electricity?

Conceptual: How do AI and robotics affect our world today and in the future?

Debatable: Are we relying on electronics too much in the world, today?

Content

- **Know:** Fundamentals of electric current and magnetism, Electro-magnetic energy and its applications; Basics of electric power generation; fundamentals of AI and robotics.
- **Understand:** The relationship between electricity and magnetism, Ohm's Law, and circuit analysis. The consequences of relying on electronics for Hawaii and the world. How the use of AI and robots can affect the future.
- **Do:** Circuits Lab, Magnetic Field Lab, Combo-Circuits Lab.

Summative Assessment Tasks

Critical Lab Skills Assessment (Crit. B/C)

Faraday Final Exam (Crit. A/D)

STEM Assessment Criteria

	Achievement Levels
A. Knowing and Understanding	0-8
B. Inquiring and Design (1 st half of Experimental/Design Cycle)	0-8
C. Processing and Evaluation (2 nd half of Experimental/Design Cycle)	0-8
D. Reflecting on the Impacts of Science/Engineering	0-8

Student activities will be assessed against the criteria listed above. End of quarter letter grades will be assigned based on student learning progress over the quarter.

Contact Information: email deachus@starofthesea.org

- Parents with questions or concerns may directly contact any instructor.
- Students are encouraged to use study hall or lunch recess to obtain additional learning support. In addition, **the Innovations Lab will be open each afternoon (except Mini-days) from 4:15 pm until 5:30 pm for any student needing help or to finish lab work.**
- Students and/or parents should use email to contact their instructor to make arrangements for additional learning opportunities.